Hisaab: a voice ledger for kirana shops that double-checks every rupee

Solution pitch for the Great Indian AI Internship Challenge 2026

The problem

A kirana owner records udhaar (credit) while serving customers, so speaking is the only practical way to write an entry. But speech recognition fails quietly on the words that matter most. In our tests "paune do sau" (₹175) came back as ₹1,750, "dedh sau" (₹150) as ₹1,500, and "सात हज़ार" (7,000) as "साठ हज़ार" (60,000). One misheard word changes a customer's balance tenfold, and nobody notices.

What Hisaab does

The shopkeeper says the entry in Hindi or Hinglish, for example "ramesh ko dhai hazaar udhaar likho". Hisaab sends the same audio to two Gnani speech models and compares the amounts they read. A small amount that both models agree on is saved straight away. Any disagreement, a missing amount, an amount under ₹10, or any amount of ₹1,000 or more is read back aloud by Timbre ("₹175 ya ₹1,750?") and needs one tap to confirm. A ledger page shows udhaar, jama and each customer's balance.

Gnani models used

Prisma v3.0 (speech to text, with number words and hint words for idioms such as dedh, dhai, sawa and paune), Prisma v2.5 (a second, independent reading in digits) and Timbre v2.5 (spoken read-backs and the synthetic test voice). Evon was not used: its open weights need more memory than a 16 GB laptop has.

Who it is for

Kirana and small-shop owners who speak Hindi or Hinglish, and developers who can reuse the amount check and the evaluation script, both open source in the repository.

What we measured

Test	Set	Result
Synthetic stress test	160 clips: 16 amounts, 2 repeats, clean, phone-band and three noise levels	Prisma v3.0 with digits: 83% right. With number words and hint words: 99%. No wrong amount was saved without a question
Real voices	30 lines recorded by our team	Prisma v3.0: 90% right (93% with hints). Prisma v2.5: 93%. No wrong amount was saved without a question

The tenfold errors came from Prisma v3.0 turning spoken numbers into digits. Asking for number words and adding hint words fixed most of them, and the second model flagged the rest.

Limits

The real-voice set is only 30 lines, and the synthetic set uses one Timbre voice. We chose the hint words and fixed spelling gaps after seeing these clips, so the scores are best-case. One mishear shared by both models (7,500 read as 60,500) was caught only because amounts of ₹1,000 and above always ask. Confirmation is by tap, not voice, and only Hindi and Hinglish amounts are handled.

Next

Spoken yes or no instead of taps, more speakers and real shop noise, Kannada and Tamil, and a Timbre voice reminder for customers with a balance.